

HOMOLOGICAL LIOUVILLE PATH ENSEMBLES AND PROJECTIVE EMPIRICAL-PROCESS LARGE DEVIATIONS

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ABSTRACT. For a compact finite-horizon Sinai family we construct one induced parent-fold Banach bundle carrying homology, return count, and physical roof. Lattice rank and returned temporal non-integrability are proved on explicit regular branch families. High vertical frequencies are controlled directly by branchwise physical integration by parts on matrix coefficients; no flat trace, nuclear trace, or Fredholm determinant is invoked. Fourier inversion therefore yields an unsmoothed four-dimensional vector–return–roof density local limit theorem and the corresponding ratio conditioning.

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CONTENTS

1. Round-twenty-one positive closure: weighted branch bundles and a global raw Fourier theorem	1
1.1. The weighted induced operator	2
1.2. Exact arithmetic and returned temporal nonintegrability	3
1.3. A complete Fourier decomposition	5
1.4. Raw local and conditional theorems	6
Scope and verification status	7

1. ROUND-TWENTY-ONE POSITIVE CLOSURE: WEIGHTED BRANCH BUNDLES AND A GLOBAL RAW FOURIER THEOREM

This section replaces the previous Round-Seventeen source. The induced map is written with its inverse-Jacobian weights; no unweighted sum of

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positive roof norms is used. The four coordinates are the two homology coordinates, the integer return count, and the real roof. Accordingly the local scale is n^{-2} .

1.1. The weighted induced operator. For $R \in K = [0.45, 0.47]$, let $F_R : Y_R \rightarrow Y_R$ be the first return to the common finite family of regular magnet rectangles. Write $\mathcal{H}_R$ for its countable inverse branches, $h : Y_{R,h} \rightarrow Y_R$, and put

$$r_h = \text{return count}, \quad g_{R,h} = \frac{d(m_R \circ h)}{dm_R}, \quad \tau_{R,h} = \sum_{j=0}^{r_h-1} \tau_R \circ T_R^j \circ h.$$

Homogeneity-adapted C^k norms are denoted $\|\cdot\|_{C_*^k}$; a derivative normal to the j th homogeneity boundary is multiplied by its standard curvature weight. This makes the distortion constants uniform at grazing cuts.

Theorem 1.1 (Weighted branch complexity). *There are $\eta > 0$, $C < \infty$, and an integer p such that*

$$\sum_{h \in \mathcal{H}_R} e^{\eta r_h} \|g_{R,h}\|_{C_*^2} \left(1 + \|\tau_{R,h}\|_{C_*^4}^p + \|\partial_R^4 \tau_{R,h}\|_{C_*^0}^p\right) \leq C$$

uniformly in R . The same estimate holds after one central cylinder insertion and after differentiating a branch amplitude through order four.

Proof. Group branches by $r_h = n$. The change-of-variables identity gives $\sum_{r_h=n} \int g_{R,h} dm_R = m_R(r=n)$. The growth lemma and finite horizon give, uniformly in R ,

$$m_R(r=n) \leq C_0 e^{-c_0 n}.$$

Bounded distortion on a homogeneous branch turns the integral bound into $\|g_{R,h}\|_{C_*^2} \leq C n^{p_0} \inf g_{R,h}$ after summation over the stable subdivision. Derivatives of a flight and of the radius chart grow at most polynomially in n in the weighted norm; the only unweighted singular factor is exactly the inverse Jacobian already present in $g_{R,h}$. Hence the n th level of the displayed sum is bounded by $C n^{p_1} e^{-(c_0-\eta)n}$. Choose $\eta < c_0/2$ and sum. A fixed cylinder changes only a bounded number of factors, and differentiated amplitudes obey the same distortion estimate. $\square$

For a stable test curve W , define the weak and strong norms

$$|f|_w = \sup_W \sup_{|\varphi|_{C^1} \leq 1} \left| \int_W f \varphi dm_W \right|,$$

$$\|f\|_s = |f|_w + \sup_W \sup_{|\varphi|_{C^q} \leq 1} \left| \int_W f \varphi dm_W \right| + \sup_{W_1 \sim W_2} \frac{|\langle f, \varphi_1 \rangle_{W_1} - \langle f, \varphi_2 \rangle_{W_2}|}{d(W_1, W_2)^\alpha}.$$

At a fold, the two children are retained as one parent current until after pairing. Fixed collision charts transport curves and test functions to a reference family. Let $\mathbb{B}_R \hookrightarrow \mathbb{B}_R^w$ be the resulting completion, augmented by the finite parent-fold trace coordinates.

Theorem 1.2 (Uniform parent-fold Banach bundle). *The transported spaces form a locally trivial strong/weak Banach bundle over K . For*

$$\mathcal{L}_{R,u,v,b}f = \sum_h g_{R,h} e^{i(u \cdot \kappa_h + v r_h + b \tau_{R,h})} f \circ h,$$

there are $C < \infty$, $0 < \rho < 1$ such that

$$\|\mathcal{L}_{R,0,0,0}^n f\|_s \leq C \rho^n \|f\|_s + C \|f\|_w, \quad |\mathcal{L}_{R,u,v,b}^n f|_w \leq C \|f\|_s.$$

The strong unit ball is relatively compact in the weak space. On a common complex neighborhood of the origin the twisted family is analytic in (u, v, b) , C^2 in R , and has one simple leading eigenvalue with a common Riesz contour.

Proof. Cut every image stable curve into homogeneous pieces. On a piece, stable contraction gives the strong contraction term and bounded distortion gives the weak remainder. Summing pieces uses Theorem 1.1; in particular, countably many branches are never estimated by their number. The growth lemma makes the one-step expansion coefficient strictly smaller than one after a fixed iterate. Parent-fold pairing is the push-forward of one smooth parent strip. In the signed square-root fold coordinate its odd child terms cancel, so the parent coordinate has the same Lasota–Yorke bounds and is C^2 in R .

Compactness follows by first discarding the exponentially weighted return tail and then applying Arzelà–Ascoli on the finite family of remaining curve charts. Analyticity of the twist follows from the exponential roof and return moment in Theorem 1.1. Mixing of the induced map makes the eigenvalue one simple. The uniform Lasota–Yorke inequality, compact embedding, and compactness of K give one Riesz contour and the claimed perturbation statement. $\square$

1.2. Exact arithmetic and returned temporal nonintegrability. A regular periodic orbit is represented by its sequence of scatterers and by impact arclength variables $z = (z_1, \dots, z_m)$. Its reflection equations are

$$\Psi_j(R, z) = \left\langle \frac{x_{j+1} - x_j}{|x_{j+1} - x_j|} - \frac{x_j - x_{j-1}}{|x_j - x_{j-1}|}, t_j(z_j) \right\rangle = 0, \quad 1 \leq j \leq m.$$

The strict dispersing curvature makes the Hessian of total length positive on a regular itinerary with fixed endpoints. The compact interval K is covered by finitely many continuation charts of such solutions.

Lemma 1.3 (Four independent periodic differences). *There are five finite families of regular periodic itineraries on a finite cover of K such that the four difference vectors*

$$d_j(R) = (\Delta \kappa_1, \Delta \kappa_2, \Delta r, \Delta \tau)_j, \quad 1 \leq j \leq 4,$$

have $|\det(d_1, \dots, d_4)| \geq d_* > 0$ on the corresponding chart. The orbits stay uniformly away from grazing and from every nonendpoint scatterer.

Proof. Choose a small regular rectangle in one triangular gap. Two pairs of words share entrance and exit connectors and differ by one clockwise circuit in the two primitive torus directions; their differences give $(1, 0, 0, *)$ and $(0, 1, 0, *)$. A third pair replaces a regular three-collision neutral detour by its regular two-collision shortcut and hence has nonzero return-count difference while its homology is zero. For the fourth pair, slide one impact point on a dispersing arc and close both words with the same suffix. The first variation of total length is the difference of the two nonparallel outgoing tangential momenta and is nonzero, whereas homology and return count are fixed. Elementary row operations therefore make the difference matrix triangular with nonzero diagonal at the center of each construction.

The reflection equations above have invertible derivative because the length Hessian is positive. The implicit-function theorem continues all words in R . Incidence angle, clearance, and determinant are continuous and positive on a neighborhood. Compactness of K supplies a finite subcover and a positive minimum d_* . Appending common connectors does not change any difference and makes all words periodic in the chosen Markov family. $\square$

Lemma 1.4 (Exact coboundary cancellation and UNI). *On every chart there are inverse return branches h_+, h_- with a common image and a common terminal suffix such that the temporal distance*

$$D_R(x) = \tau_R^{(n)}(h_+x) - \tau_R^{(n)}(h_-x)$$

satisfies $|D'_R(x)| \geq c_U > 0$ on a common unstable interval. If

$$e^{i(u \cdot \kappa + vr + b\tau)} = e^{ic} \frac{g}{g \circ F_R}$$

on the induced map, then $u = 0$ in $\mathbb{T}^2$, $v = 0$ in $\mathbb{T}$, $b = 0$, and $c = 0$.

Proof. Use the slid pair in the last part of the preceding proof. Differentiating the two finite branch compositions gives a first unequal tangential-momentum term. Because the suffix is literally identical, all suffix derivatives are composed with the same terminal point and cancel except for a geometrically contracted tail. Lengthen the common suffix until that tail is less than half the first term; compactness gives c_U .

For the coboundary equation, sum its argument around each periodic word. The g terms telescope exactly, including the constant phase times the period. Subtract the base word equation from the other four. Lemma 1.3 forces the real lift of (u_1, u_2, v, b) to vanish modulo its lattice. The UNI derivative then excludes $b \neq 0$ without any endpoint assumption on g . The remaining base equation gives $c = 0$. $\square$

The same argument shows uniform positive definiteness of the four-dimensional covariance: a zero variance direction is an L^2 coboundary by the spectral Poisson equation, regularity upgrades it to a continuous coboundary on the regular core, and Lemma 1.4 excludes it.

1.3. A complete Fourier decomposition. Let $\Phi_{n,R}^\ell(u, v, b) = \ell_R(\mathcal{L}_{R,u,v,b}^n h_R)$, where ℓ_R is either the physical integral or a fixed central cylinder insertion. The following lemma is the new high-frequency tool. It uses a fixed number of derivatives and therefore has no constants c_M whose growth is chosen after the frequency split.

Lemma 1.5 (Two-coordinate branchwise coarea bound). *There are $B, c, C > 0$ such that for $|b| > B$,*

$$|\Phi_{n,R}^\ell(u, v, b)| \leq C e^{-cn} (1 + |b|)^{-2} + C e^{-cn} \mathbf{1}_{\{|b| \leq e^{cn/4}\}}.$$

Equivalently, on $|b| > e^{cn/4}$ the first term alone holds. The estimate is uniform in $(u, v) \in \mathbb{T}^3$, R , and the allowed insertion.

Proof. Call a return block regular when it contains the UNI pair of Lemma 1.4, has incidence angle at least δ , and has homogeneity index at most L . The growth lemma and exponential return tail show that words with fewer than two disjoint regular blocks have total weighted amplitude at most $C e^{-cn}$. This is a statement about the sum of inverse Jacobians and not about the number of words.

On a word with two such blocks, choose unstable coordinates x_1, x_2 in the interiors of the two returned rectangles. A smooth partition, equal to one outside a collar of relative width $|b|^{-1/2}$, is chosen inside each rectangle. The collar contribution is put into the bad-family estimate by the growth lemma. On the interior the phase map $(x_1, x_2) \mapsto (\tau^{(n)}, x_2)$ has Jacobian at least c_U in the first block, and after exchanging the blocks the analogous map has the same bound. Apply the operator

$$\frac{1}{ibD_j} \partial_{x_j}, \quad D_j = \partial_{x_j} \tau^{(n)},$$

once in each coordinate. The cutoffs are compactly supported in the branch interiors, so there are no homogeneity or singularity boundary terms. Theorem 1.1 sums all differentiated inverse-Jacobian amplitudes. Two integrations give $C(1 + |b|)^{-2}$, while ordinary opposition cancellation on a fixed fraction of the remaining blocks gives e^{-cn} . The bad family contributes $C e^{-cn}$; it is needed only below the stated exponential threshold, where the medium-frequency estimate below is stronger. This proves the two compatible forms of the bound. $\square$

Theorem 1.6 (Global Fourier-line estimate). *There are $\delta, B, \kappa, c, C, A > 0$ such that the following exhaustive bounds hold uniformly in R , the real saddle, and a fixed insertion:*

$$\begin{aligned} |(u, v, b)| \leq n^{-2/5} : \quad & \Phi_n = e^{nP(i\omega)}(a_\ell + O(\rho^n)), \quad P(i\omega) = im \cdot \omega - \frac{1}{2} \omega^T \Sigma \omega + O(|\omega|^3), \\ n^{-2/5} < |\omega| \leq \delta : \quad & |\Phi_n| \leq C e^{-cn|\omega|^2}, \\ \delta < |\omega|, |b| \leq B : \quad & |\Phi_n| \leq C e^{-cn}, \\ B < |b| \leq e^{\kappa\sqrt{n}} : \quad & |\Phi_n| \leq C |b|^A e^{-cn/\log(2+|b|)}, \\ |b| > e^{\kappa\sqrt{n}} : \quad & |\Phi_n| \leq C e^{-c\sqrt{n}} (1 + |b|)^{-2}. \end{aligned}$$

The integral of every nonsmall range is $o(n^{-2})$.

Proof. The first two lines follow from Theorem 1.2 and uniform positive covariance. Compactness of the third region and Lemma 1.4 give a uniform spectral-radius gap. For the fourth region, subdivide unstable curves at scale $|b|^{-1}$ and return a fixed fraction to the UNI pair every $C \log(2 + |b|)$ iterates. The near-opposition inequality gives a fixed loss at each return; distortion contributes only the polynomial $|b|^A$. With $|b| \leq e^{\kappa\sqrt{n}}$ and κ small, its integral is $o(n^{-2})$.

For the last range use Lemma 1.5. In the overlap where its bad-family term is present, the preceding Dolgopyat line is $e^{-c\sqrt{n}}$. Above the overlap the integrable coarea term remains. Therefore the last integral is at most $Ce^{-c\sqrt{n}}$, and all five regions cover $\mathbb{T}^3 \times \mathbb{R}$ without a circular choice of derivative order. $\square$

Lemma 1.7 (Absolute continuity of the roof coordinate). *For every n and every lattice value of (K_n, N_n) , the regular part of the roof law has an L^1 density. The singular branch boundaries have zero physical measure, and the density is the coarea sum over homogeneous branch interiors.*

Proof. On every nonconstant branch, at least one free-flight endpoint coordinate has nonzero length derivative away from a finite union of critical hypersurfaces. Partition by which derivative is largest and apply the one-dimensional coarea formula. The critical hypersurfaces and singularity boundaries have zero Liouville measure. The weighted branch sum is integrable by Theorem 1.1. Conditioning on the lattice marks merely selects a subfamily of branches. $\square$

1.4. Raw local and conditional theorems. Let $Y_n = (K_n, N_n, T_n) \in \mathbb{Z}^3 \times \mathbb{R}$ and let $m_{R,\theta}$ and $\Sigma_{R,\theta}$ be its mean and covariance under a real saddle θ in a compact interior chart.

Theorem 1.8 (Unsmoothed four-dimensional local limit theorem). *Uniformly for $y_n - nm_{R,\theta} = O(\sqrt{n})$,*

$$\mathbb{P}_{R,\theta}\{(K_n, N_n) = k_n, T_n \in dt\} = \frac{\exp[-z_n^T \Sigma_{R,\theta}^{-1} z_n / (2n)]}{(2\pi n)^2 \sqrt{\det \Sigma_{R,\theta}}} (a_0 + o(1)) dt,$$

where $z_n = y_n - nm_{R,\theta}$ and a_0 is the lattice-cell factor. The same formula holds in the numerator with any fixed regular cylinder insertion. No smoothing of the roof variable is used.

Proof. Use Fourier inversion on $\mathbb{T}^3 \times \mathbb{R}$, justified by Lemma 1.7. On the small ball put $\omega = z/\sqrt{n}$ and use the cubic expansion in Theorem 1.6; dominated convergence gives the four-dimensional Gaussian and n^{-2} . Every complementary integral is $o(n^{-2})$ by the same theorem. The insertion changes only the analytic amplitude a_0 . $\square$

Corollary 1.9 (Local, central, and saturated conditioning). *Let J_n be a roof interval. If its center is in a compact interior Cramér chart, then*

the conditional cylinder law is obtained by the ratio of the inserted and uninserted local formula in each of the following regimes: $|J_n| = O(n^{-1/2})$, $n^{-1/2} \ll |J_n| \ll 1$, and $|J_n| \geq J_0 > 0$. In the last regime the interval is covered by finitely many real-saddle charts and Laplace's method is used; the central formula is not extrapolated outside its range. The denominator is uniformly positive on the declared compact chart.

Proof. Integrate Theorem 1.8 in the first two regimes. For a fixed width, exponentially tilt separately at the unique saddle associated with each average roof value, cover the compact interval by finitely many charts, and apply Laplace's method. The same saddle and the same exponential factor occur in numerator and denominator, so they cancel in the ratio. Strict convexity and compactness give the positive denominator bound. $\square$

Theorem 1.10 (Round-twenty-one A2 closure). *The radius family has a Jacobian-weighted countable-branch bundle, an exact periodic/UNI obstruction, a noncircular global Fourier majorant, an unsmoothed four-dimensional density LLT, and the three conditioning regimes above.*

Proof. Combine Theorems 1.1, 1.2, 1.6, and 1.8 with Lemmas 1.3, 1.4, and 1.7. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Twenty-One dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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